

AHMED MAGDY MOHAMED

Mobile Applications Engineer | Flutter & Native Android

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PROFESSIONAL SUMMARY

Mobile Applications Engineer focused on Flutter and native Android development with Kotlin and Dart. Experience building cross-platform and native mobile applications using Jetpack Compose, Firebase, Cloud Firestore, and REST APIs (Retrofit), with a working understanding of Clean Architecture, BLoC/Cubit, and MVVM patterns. Technical foundation developed through intensive training programs (DEPI, ITI), mobile-focused internships, and independent projects, sensor integration, and unit testing, supported by a full-stack background in web and backend technologies.

TECHNICAL SKILLS

• Mobile Development

- Flutter, Dart
- BLoC, Cubit
- Clean Architecture
- Firebase, Cloud Firestore
- Firebase Authentication
- REST APIs

• Native Android

- Kotlin, Java
- Jetpack Compose, Material Design
- Android SDK, XML, ViewBinding
- Retrofit, Navigation

• Backend & APIs

- ASP.NET Core, C#
- ASP.NET Core Web API
- Entity Framework Core
- Node.js, Express.js
- Spring Boot, Spring JPA
- REST APIs, SignalR, JWT

• Databases

- SQLite
- Room
- Cloud Firestore

• Testing & Tools

- Unit Testing, Functional Testing
- Git, GitHub, Android Studio, VS Code, Postman

• Additional Technologies

- Angular, React, TypeScript
- JavaScript, HTML5, CSS3
- Tailwind CSS, Bootstrap 5

EXPERIENCE

- [Digital Egypt Pioneers Initiative \(DEPI\)](#) — [Global Knowledge](#) [Mar 2026 – Aug 2026]
 - Completed a 6-month DEPI training program (Ministry of Communications and Information Technology, in partnership with Global Knowledge) focused on Flutter and cross-platform mobile development.
 - Built advanced skills in Dart and Flutter, developing, testing, and deploying responsive mobile applications with modern app architecture.
- [Information Technology Institute \(ITI\)](#) [Dec 2025 – Feb 2026]
 - Completed Flutter and Dart training program covering fundamental through mobile application concepts.
 - Built interactive, responsive UIs applying core architecture principles, culminating in a Tic-Tac-Toe (XO) graduation project.
- [Digital Egypt Pioneers Initiative \(DEPI\)](#) — [YAT Learning Center](#) | [Certificate ↗](#) [Jun 2025 – Dec 2025]
 - Completed a 6-month DEPI training program at YAT Center focused on Kotlin and native Android application
 - Gained hands-on experience designing, developing, and testing Android applications using Android Studio, Java, and Kotlin, alongside teamwork and communication skills.
- [CodeAlpha](#) [Mar 2025 – May 2025]
 - Kotlin-focused mobile development training, applying core Android and Kotlin concepts through hands-on, mentor-guided project work in a structured internship program.

- [Decode Labs](#)

[Sep 2026 – Nov 2026]

- Flutter-focused hands-on mobile development experience through project-based learning, guided mentorship, and portfolio-oriented practical work.

PROJECTS

- [Mobile Discovery App ↗](#) — **Android Engineer, Architecture, Networking & Testing**

Technologies: Kotlin, Jetpack Compose, Material Design 3, Retrofit, Navigation Component, JUnit

- Developed a reactive, declarative UI with Jetpack Compose as the DEPI Android graduation project.
- Architected a Retrofit networking layer with API credentials secured through a Gradle secrets plugin.
- Authored a unit test suite achieving a 100% pass rate across mapper, ViewModel, and repository tests.

- [Pattern & Sign Language ↗](#) — **Flutter Engineer, EdTech & Accessibility (Freelance)** | [Google Play ↗](#)

Technologies: Flutter, Dart, Firebase, YouTube Embedded API, RTL UI

- Built platform helping the deaf and hard-of-hearing community learn garment drafting.
- Designed an interactive evaluation engine with pre- and post-tests to measure skill retention.
- Shipped responsive RTL layouts with a Firebase-backed community feedback board and embedded instructional video content.

- [Desert Maps & Offline GPS ↗](#) — **Lead Android Engineer (Freelance)** | [Google Play ↗](#)

Technologies: Kotlin, Android Location API, Magnetometer, Orientation Sensors, Gradle Kotlin DSL

- Built a fully offline GPS land-navigation system for desert exploration without cellular coverage.
- Integrated hardware magnetometer and orientation sensors for compass navigation and Qibla direction.
- Implemented coordinate parsing, distance and altitude calculations, and offline waypoint categorization.

- [Online XO ↗](#) — **Mobile Architect & Flutter Engineer (ITI Graduation Project)**

Technologies: Flutter, Dart, flutter_bloc/Cubit, Equatable, Firebase Auth, Cloud Firestore

- Delivered real-time remote board updates for a multiplayer game using Firestore stream listeners.
- Separated presentation, Cubit state machines, domain models, and infrastructure under Clean Architecture.
- Built public/private matchmaking lobbies with 30-second turn timers and atomic score updates via Firestore transactions.

- [HealthyMe ↗](#) — **Mobile Developer (Freelance)**

Technologies: Kotlin, Android SDK, ViewBinding, DecelerateInterpolator, GradientDrawable

- Built dynamic theming that adapts to BMI category: underweight, normal, overweight, or obese.
- Designed entrance transitions using custom interpolators and programmatic gradient styling.

- [Prayer App \(Athan & Qibla\) ↗](#) — **Mobile Engineer, Sensors & Networking**

Technologies: Kotlin, Retrofit 2, Gson, SensorManager, FusedLocationProviderClient

- Calculated precise Kaaba bearing using Haversine/great-circle formulas with atan2, sin, and cos.
- Fused magnetometer and accelerometer input for a smooth, stable rotating compass.
- Integrated the Aladhan REST API for dynamic prayer times with asynchronous parsing and Azkar state cycling.

EDUCATION

B.Sc. in Computer Science & Artificial Intelligence, Information Systems

Faculty of Computers and Artificial Intelligence, Helwan University | 2023 – 2027 | GPA: 3.63